

# Theorems of Dirac & Ore

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# Introduction

The Theorems were given by a Hungarian-British Mathematician G.A. Dirac (1952) and Norwegian Mathematician O. Ore (1960), the former being the deduction of the latter.

The Theorems give sufficient conditions for a graph to be Hamiltonian.



**G.A. Dirac**



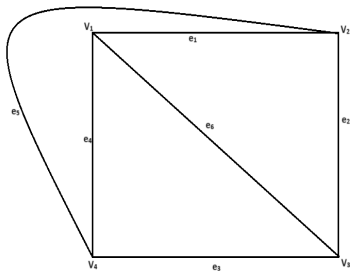
**O.Ore**

# Hamiltonian Path

## Hamiltonian Path

A path in a graph  $G$  which contains all the vertices of  $G$  is called a Hamiltonian Path.

**Example:** Here the Hamiltonian path is  $(V_2, e_1, V_1, e_6, V_3, e_3, V_4)$ .



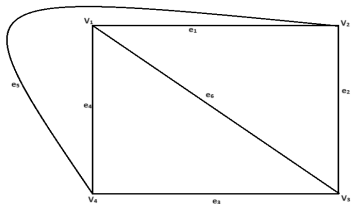
**Graph with Hamiltonian Path**

# Hamiltonian Cycle & Hamiltonian Graph

## Hamiltonian Cycle & Hamiltonian Graph

A cycle in a graph  $G$  which contains every vertex of  $G$  is called a Hamiltonian cycle. A graph having a Hamiltonian cycle is called a Hamiltonian Graph.

**Example:** Here the Hamiltonian Cycle is  $(V_1, e_1, V_2, e_2, V_3, e_3, V_4, e_4, V_1)$ . So the following Graph is Hamiltonian.



**Graph with Hamiltonian Cycle**

# Ore's Theorem

## Statement

Let  $G$  be a simple  $n$ -vertex graph, where  $n \geq 3$ , such that  $\deg(x) + \deg(y) \geq n$  for each pair of non-adjacent vertices  $x$  and  $y$ . Then  $G$  is Hamiltonian.

## Proof

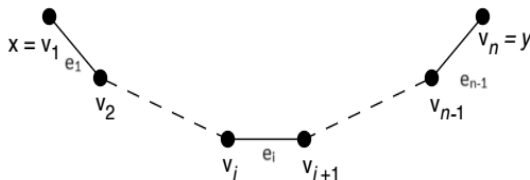
- We prove it by method of contradiction.
- Assume that the theorem is false and let  $G$  be a maximal counter-example.\*

\*The proof is taken from "Gross, Jonathan L., and Jay Yellen. Graph theory and its applications. CRC press, 2005."

# Proof of Ore's Theorem

- Let  $x$  and  $y$  be two non-adjacent vertices of  $G$ . It suffices to show that  $\deg(x) + \deg(y) \leq n - 1$ .
- Since  $G + xy$  contains a Hamiltonian cycle,  $G$  have a Hamiltonian path whose end points are  $x$  and  $y$ .

Let  $(x = v_1, e_1, v_2, e_2, \dots, v_i, e_i, \dots, v_{n-1}, e_{n-1}, v_n = y)$  be such path.

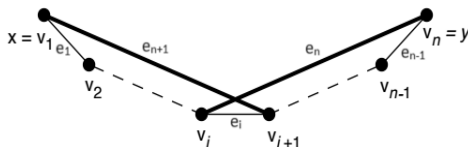


A Hamiltonian path in  $G$

# Proof of Ore's Theorem (Continued)

**Claim:** For each  $i = 2, \dots, n-1$  at least one of the pairs  $v_1, v_{i+1}$  and  $v_i, v_n$  is non-adjacent.

Otherwise  $(v_1, e_1, v_2, \dots, v_i, e_n, v_n, e_{n-1}, v_{n-1}, \dots, v_{i+1}, e_{n+1}, v_1)$  would be a Hamiltonian cycle in  $G$ .



Hamiltonian Cycle in  $G$



# Proof of Ore's Theorem (Continued)

- It means if  $(a_{i,j})$  is the adjacency matrix for  $G$ , then

$$a_{1,i+1} + a_{i,n} \leq 1 \text{ for } i = 2, \dots, n-2.$$

- Therefore,  $\deg(x) + \deg(y) = \sum_{i=2}^{n-1} a_{1,i} + \sum_{i=2}^{n-1} a_{i,n}$ 
$$\begin{aligned} &= a_{1,2} + \sum_{i=3}^{n-1} a_{1,i} + \sum_{i=2}^{n-2} a_{i,n} + a_{n-1,n} \\ &= 1 + \sum_{i=2}^{n-2} a_{1,i+1} + \sum_{i=2}^{n-2} a_{i,n} + 1 \\ &= 2 + \sum_{i=2}^{n-2} (a_{1,i+1} + a_{i,n}) \\ &\leq 2 + n - 3 = n - 1 \end{aligned}$$

Which is the required contradiction (to the given condition of the theorem).

# Dirac's Theorem

## Statement

Let  $G$  be a simple  $n$ -vertex graph, where  $n \geq 3$ , such that  $\deg(v) \geq \frac{n}{2}$  for each vertex  $v$ . Then  $G$  is Hamiltonian.

## Proof

- It follows immediately from the Ore's Theorem.

- Gross, Jonathan L., and Jay Yellen. Graph theory and its applications. CRC press, 2005.
- West, Douglas Brent. Introduction to graph theory. Vol. 2. Upper Saddle River: Prentice hall, 2001.
- Malik, D. S., and Mridul K. Sen. "Discrete mathematical structures: theory and application." (No Title) (2004).

*Thank You!*